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IONIC THRUSTER METHODS AND APPARATUS FOR AIRCRAFT

Abstract

Ionic thruster methods and apparatus for aircraft are disclosed. An example ionic thruster for aircraft includes a nozzle. The nozzle includes an outlet and an inlet, the inlet to receive fluid and containing an electrode mount. A ground electrode is disposed within the nozzle. Conducting pins are coupled to the electrode mount, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end, the pins spaced apart from the ground electrode.

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Background/Summary

FIELD

[0001] This disclosure relates generally to aircraft and, more particularly, to ionic thruster methods and apparatus for aircraft.

BACKGROUND

[0002] In recent years, ionic wind has been studied as a mechanism to induce fluid flow. A high voltage electric field between two electrodes can cause air between the electrodes to ionize, which causes an electrical current in the form of a corona discharge. Moving particles within the corona discharge collide with neutral particles in the air, inducing movement of the surrounding air. Thus, a prolonged or sustained corona discharge creates airflow that is often called ionic wind. Ionic wind can be utilized in aircraft to generate thrust. Currently ionic wind thrusters utilize wire electrodes on or near leading edges of wings.

SUMMARY

[0003] An example ionic thruster for aircraft described herein includes a nozzle including an outlet and an inlet, the inlet to receive fluid and containing an electrode mount, a ground electrode disposed within the nozzle, and conducting pins coupled to the electrode mount, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end, the pins spaced apart from the ground electrode.

[0004] An example aircraft described herein includes a voltage source and a thruster including a body. The body includes an outlet and an inlet, the body to receive fluid and containing an electrode mount, a ground electrode disposed within the body and electrically coupled to the voltage source, and conducting pins coupled to the electrode mount and electrically coupled to the voltage source, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end.

[0005] An example method for generating thrust on an aircraft is described herein. The method includes providing a voltage to a thruster. The thruster including a nozzle including an outlet and an inlet, the inlet to receive air, an electrode mount disposed within the inlet, a ground electrode disposed within the nozzle, and conducting pins coupled to the electrode mount, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end, the pins spaced apart from the ground electrode. The method includes generating a corona discharge with the voltage, the corona discharge to extend between the conducting pins and the ground electrode, and inducing an ionic wind with the corona discharge, the ionic wind to generate the thrust.

[0006] The features, functions, and advantages of that have been discussed can be achieved independently in various embodiments or may be combined in yet other embodiments further details of which can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is an example aircraft on which an example ionic thruster can be implemented.

[0008] FIGS. 2A and 2B are cross-section illustrations of the example ionic thruster of FIG. 1.

[0009] FIG. 3 is a cross-section illustration of an example ionic thruster that includes an example converging-diverging nozzle.

[0010] FIG. 4 illustrates example corona discharges and example ionic wind resulting from operation of the example ionic thruster of FIGS. 2A-2B.

[0011] FIG. 5 illustrates example corona discharges and example ionic wind resulting from operation of an example ionic thruster that includes an example electromagnet to guide the corona discharges.

[0012] FIGS. 6A and 6B are cross-section illustrations of an example ionic thruster that include an example dielectric guide.

[0013] FIG. 7 illustrates the example ionic thruster of FIGS. 6A-6B with an example dielectric guide that includes example converging-diverging geometry.

[0014] FIGS. **8A** and **8B** illustrate a front and a side view of an example arrangement of the example ionic thrusters of FIG. **1** to form a larger ionic thruster.

[0015] FIG. **9** is a flowchart describing an example method of generating and controlling thrust for an aircraft using an ionic thruster.

[0016] In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not necessarily to scale. Instead, the thickness of the layers or regions may be enlarged in the drawings. Although the figures show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular.

DETAILED DESCRIPTION

[0017] Known ionic wind thrusters for aircraft utilize wires as

[0018] electrodes to generate an electric field. While a wire electrode allows for an even distribution of an electric field across a long surface, such as the leading edge of a lift surface, the electric field is limited in strength. An electric field strength can be increased by using an electrode geometry that ends in a sharp surface, such as a blade or a tip of a pin. Example ionic thrusters described herein utilize pin shaped electrodes to generate stronger electric fields and corresponding stronger thrusts than current ionic wind thrusters. Known ionic wind thrusters are required to be placed in front of surfaces such as wings to generate thrust. In contrast, ionic thrusters described herein can be fixed to an aircraft in any advantageous position or configuration, such as under a wing or on a tail. Thus, the ionic thrusters described herein can be easily integrated into many existing aircraft designs.

[0019] FIG. **1** is an example aircraft **100** on which an example ionic thruster **102** can be implemented. The example aircraft **100** is an unmanned aircraft with example wings **104** and an example tail **106**. The ionic thrusters **102** are mounted under the wings **104**. In some examples, the ionic thrusters **102** can be mounted in a different location (e.g., a different position under the wing, over the wing, on the tail, etc.). In some examples, the aircraft **100** can be a different kind of aircraft (e.g., drone, balloon, loitering aircraft, etc.). The example ionic thrusters **102** are illustrated with a cylindrical shape. In some examples, the ionic thrusters **102** can have a different shape. In some examples, the ionic thrusters **102** are surrounded by an aerodynamic surface (e.g., fairings). In other examples, the ionic thrusters **102** are mounted in nacelles extending from the aircraft **100**. In some examples, the ionic thrusters **102** are movably mounted (e.g., mounted on a gimbal) on the aircraft. The example aircraft **100** of FIG. **1** also includes an example voltage source **108** (e.g., battery, fuel cell, generator, solar panel, capacitor, etc.) to provide voltage to the ionic thrusters **102**. As described in more detail in reference to FIG. **4**, the ionic thrusters **102** are powered by the voltage source **108** to produce thrust for the aircraft **100**. An example controller **110** connects to systems of the aircraft **100** (e.g., the voltage source **108**, the thrusters **102**, etc.) to at least direct the systems' functions and modify an amount of power (e.g., voltage, current, etc.) being supplied to the systems. The controller **110** can be implemented by any type(s) and/or any number(s) of semiconductor device(s) (e.g., microprocessor(s), microcontrollers(s), etc.) and/or circuit(s).

[0020] FIGS. **2A** and **2B** are cross-section illustrations of the example ionic thruster **102** of FIG. **1**. The ionic thruster **102** includes an example nozzle **200**, an example electrode mount **202**, and an example ground electrode **204**. The nozzle **200** directs fluid (e.g., air, atmosphere, etc.) to flow through the ionic thruster **102**. The nozzle **200** has an example inlet **206** to receive fluid and an example outlet **208** to eject thrust generating fluid. In this way, the outlet **208** directs the fluid and the resulting thrust in a direction relative to the example aircraft **100**.

[0021] The electrode mount **202** of the example ionic thruster **102** of FIGS. **2A-2B** is coupled to the inlet **206** of the nozzle **200**. Example pins **210** (e.g., conducting pins, electrode pins, etc.) are coupled to the electrode mount **202** to support and orient the pins **210** relative to the ground electrode **204**. The pins are aligned such that an end of a pin **210** is closer to the ground electrode

204 than an opposite end of the pin **210**. In some examples, the pins **210** are oriented parallel to a central axis of the nozzle **200** between the inlet **206** and the outlet **208**. In other examples, the pins **210** have a different alignment (e.g., perpendicular to the ground electrode **204**, skewed from the central axis, etc.). The pins **210** are cylindrical in shape to facilitate generating an electric field at the tips of the pins **210**. In other examples, the pins **210** can have different shapes (e.g., pointed tips, blades, sharp rings, etc.) to facilitate electric field generation. In some examples, the pins **210** are connected to the voltage source **108** through a series of conducting wires. In other examples, the electrode mount **202** is composed of a conductive material and the pins **210** are connected the voltage source **108** through the electrode mount **202**.

[0022] The electrode mount **202** is shaped to allow fluid to flow through the nozzle **200**. As such, the electrode mount **202** has holes or openings that allow fluids to flow through. The illustrated example of FIG. 2A shows the electrode mount **202** with a plurality of example radial supports **212** extending away from a central portion of the electrode mount **202** towards an internal wall **215** of the nozzle **200** (e.g., an internal wall of the inlet **206**). In other examples, the electrode mount **202** can have any other shape that allows fluid to flow through the nozzle **200**. In some examples, the electrode mount **202** is shaped to support an advantageous arrangement of the pins **210**. In some examples, the electrode mount **202** is part of the nozzle **200** shaped to hold the pins **210**.

[0023] In the ionic thruster **102** of FIGS. 2A-2B, the ground electrode **204** is coupled inside the nozzle **200**. The ground electrode **204** is spaced apart from the pins **210** to allow for an electric field to form between the pins **210** and the ground electrode **204**. FIGS. 2A-2B show the ground electrode **204** contacting and/or coupled to an example spacer **214**. The spacer **214** maintains an example spacing between the pins **210** and the ground electrode **204**. In other examples, the spacer **214** can be a different size. In some examples, the spacer **214** is not a separate component, but a geometric feature of the nozzle **200** that accepts the ground electrode **204**. The example ground electrode **204** of FIG. 2A is a wire mesh that has example openings **216** to allow fluid to flow past the ground electrode **204** towards the outlet **208** of the nozzle **200**. In some examples, the ground electrode **204** is a plate with holes that allow fluid to flow from the inlet **206** to the outlet **208**. The holes can have any advantageous size or shape (e.g., circular, polygonal, etc.). The example ground electrode **204** is shown in FIG. 2A with an example thickness, but other example ground electrodes **204** can have a larger or smaller thickness. The ground electrode **204** is electrically coupled to the voltage source **108**. In some examples, the nozzle **200** is composed of a non-conductive material (e.g., insulating material) to prevent current flow between the ground electrode **204**, the pins **210**, and/or the electrode mount **202**. The nozzle **200** extends past the ground electrode **204**, ending at the outlet **208** after an example distance. In other examples, the nozzle **200** extends a different distance (e.g., is shorter or longer). In some examples, the outlet **208** is coincident with the ground electrode **204** and the nozzle **200** does not extend past the ground electrode **204**.

[0024] FIG. 3 is a cross-section illustration of an example ionic thruster **300** that includes an example converging-diverging nozzle **302**. The nozzle **302** has an example inlet **304**, an example outlet **306**, an example thrust chamber **308** (housing the electrode mount **202**, the pins **210**, and the ground electrode **204**). The inlet **304** couples to the thrust chamber **308** such that fluid flows from the inlet **304** into the thrust chamber **308**. In some examples, the electrode mount **202** is coupled to the inside of the thrust chamber **308**, adjacent to the inlet **304**. The pins **210** are coupled to (e.g., embedded in, welded to, or otherwise fixed to) the electrode mount **202**. The pins **210** extend from the electrode mount **202** towards the ground electrode **204**. The ground electrode **204** is coupled to the inside of the thrust chamber **308**, opposite the electrode mount **202**. The outlet **306** couples to the thrust chamber **308** opposite the inlet **304**, such that fluid flows from the thrust chamber **308** into the outlet **306**. In some examples, the ground electrode **204** is coupled to the inside of the thrust chamber **308** adjacent to the outlet **306**. The inlet **304** and the outlet **306** have larger cross-sectional areas (e.g., diameters, openings, etc.) at the outermost ends that reduce or converge to meet the smaller cross-sectional area of the thrust chamber **308**. In other words, the nozzle **302**

converges from the inlet **304** to the thrust chamber **308**, and the nozzle **302** diverges from the thrust chamber **308** to the outlet **306**. In this way, the inlet **304** speeds up fluid as it enters the thrust chamber **308** and back pressure is reduced as the fluid exits the outlet **306**. At high altitudes, the inlet **304** compresses fluid (e.g., air) to facilitate generating a corona discharge and reduce the chance of an arc forming between the pins **210** and the ground electrode **204**. The example converging-diverging nozzle **302** has an example geometry including an inlet length and diameter (e.g., cross-sectional area); an example outlet length and diameter (e.g., cross-sectional area); and an example thrust chamber length and diameter (e.g., cross-sectional area). Other examples can have example inlets **304**, outlets **306**, and thrust chambers **308** with different lengths, diameters (e.g., cross-sectional areas), and/or geometries. In some examples, the inlet **304** and/or the outlet **306** increase in area linearly (e.g., conically). In other examples, the inlet **304** and/or the outlet **306** have non-linear increases in area (e.g., curved, parabolic, stepwise increases, etc.) In some examples, the cross-sections are not circular (e.g., polygonal, etc.). In some examples, the cross-sections of the inlet **304**, thrust chamber **308**, and the outlet **306** are coaxial with a central axis. In other examples, the cross-sections of the inlet **304**, the thrust chamber **308**, and the outlet **306** do not share a common central axis (i.e., are not coaxial). In some examples, the inlet **304** converges and the outlet **306** has a constant diameter. In other examples, the inlet **304** has a constant diameter and the outlet **306** diverges.

[0025] FIG. 4 illustrates example corona discharges **400** and example ionic wind **402** resulting from operation of the example ionic thruster **102** of FIGS. 2A-2B. An example fluid **404** (e.g., atmosphere, air, etc.) enters the ionic thruster **102** at the inlet **206**. In some examples, the fluid **404** enters the ionic thruster **102** at a pressure determined by the ambient environment. The pins **210** receive a voltage from the voltage source **108** (not shown). In some examples, the voltage provided to the pins **210** is high (e.g., 10 kilovolts, 20 kilovolts, etc.). In some examples the voltage can vary based on flight conditions (e.g., altitude, air speed, etc.) of the aircraft **100** of FIG. 1. In some examples, the voltage source **108** provides rapidly pulsed voltage to the pins **210**. The voltage difference between the pins **210** and the ground electrode **204** generates an electric field. The electric field, in turn, causes corona discharges **400** between the pins **210** and the ground electrode **204**. The corona discharges **400** are composed of ionized particles of the fluid **404** that move towards the ground electrode **204**. Neutral particles of the fluid **404** are accelerated by the corona discharges **400** and flow through ground electrode **204**, creating ionic wind **402** (e.g., ion wind, corona wind, electric wind, etc.) that exits the nozzle **200** at the outlet **208**. The corona discharges **400** and ionic wind **402** cause a thrust reaction in the ionic thruster **102**, which allows the ionic thruster **102** to generate thrust for the aircraft **100** of FIG. 1.

[0026] FIG. 5 illustrates example corona discharges **400** and example ionic wind **402** resulting from operation of the example ionic thruster **102** that includes an example electromagnet **500** to guide the corona discharges **400**. The electromagnet **500** is positioned to surround the corona discharges **400**. In other words, the electromagnet **500** surrounds at least the space between the pins **210** and the ground electrode **204**. In some examples, the electromagnet **500** extends beyond the pins **210** and the ground electrode **204**. FIG. 5 shows the electromagnet **500** in contact (e.g., surrounding) with the nozzle **200**. In other examples, the electromagnet **500** can be embedded or otherwise located within the nozzle **200** or the spacer **214**. Power is supplied to the electromagnet **500** to generate a magnetic field. The magnetic field guides (e.g., constricts or converges, etc.) the corona discharges **400** towards the center (e.g., a central axis) of the nozzle **200** and/or the center of the electromagnet **500**. In this way, the magnetic field generated by the electromagnet **500** acts as a converging nozzle for the fluid ions that make up the corona discharges **400** to increase a velocity of the corona discharges **400** and the resulting ionic wind **402**. In some examples, the operation of the electromagnet **500** can be varied continuously from full power to no power (e.g., a deactivated state). In this way, the corona discharges **400** and the resulting thrust generated by the ionic thruster **102** can be modified based on modifying the power supplied to the electromagnet **500**. In some

examples, the electromagnet **500** is powered with non-continuous voltage (e.g., pulsed voltage, pulse width modulated voltage, cyclic voltage, etc.). In some examples, the electromagnet **500** is a permanent magnet that does not receive power but continually generates a magnetic field.

[0027] FIGS. **6A** and **6B** are cross-section illustrations of an example ionic thruster **600** that includes an example dielectric guide **602**. The dielectric guide **602** surrounds the pins **604** while leaving channels (e.g., holes) for fluid to flow through the dielectric guide **602**. The dielectric guide **602** extends past the example pins **604** towards an example ground electrode **606**. In this way, the dielectric guide **602** isolates the pins **604** so that the electric field and/or corona discharge originating from a pin **604** interferes less with the electric field and/or corona discharge originating from a different nearby pin **604**. Therefore, a higher density (e.g., number) of pins **604** can generate corona discharges in a smaller area, making the generation of ionic wind and thrust more efficient. In some examples, the dielectric guide **602** is spaced apart from an example electrode mount **608**. In other examples, the dielectric guide **602** is in contact with the electrode mount **608**. In some examples, the dielectric guide **602** is coupled to an interior surface of an example nozzle **610** of the ionic thruster **600**. In other examples, the dielectric guide **602** and the nozzle **610** are one piece. The example ionic thruster **600** of FIGS. **6A-6B** is similar to the example ionic thruster **102** of FIGS. **2A-2B** with an addition of the dielectric guide **602**. As such, it should be understood that some components of the ionic thrusters **102**, **600** are similar and/or interchangeable (e.g., the ground electrodes **204**, **606**, the nozzles **200**, **610**, the pins **210**, **604**, etc.).

[0028] FIG. **7** illustrates the example ionic thruster **600** of FIGS. **6A-6B** with an example dielectric guide **700** that includes an example converging-diverging geometry **702**. The converging-diverging geometry **702** provides holes or passages that converge (e.g., shrink in diameter) towards the tips of the pins **604** from one end of the hole, and then diverge (e.g., grow in diameter) towards an opposite end of the hole. The converging-diverging geometry **702** of the dielectric guide **700** causes fluid pressure to increase as it approaches the end of the respective pin **604**. Similar to the converging-diverging nozzle **302** of FIG. **3**, the increased pressure from the converging-diverging geometry **702** causes an increased generation of thrust in the ionic thruster **600**. Additionally, the increased pressure at the end of the pin **604** (e.g., the point at which a corona discharge is initiated) facilitates corona generation and lessens frozen flow losses in thrust generation. In some examples, the dielectric guide **700** is coupled to an interior surface of an example nozzle **610** of the ionic thruster **600**. In other examples, the dielectric guide **700** and the nozzle **610** are one piece.

[0029] FIGS. **8A** and **8B** illustrate a front and a side view of an

[0030] example arrangement **800** of the example ionic thrusters **102** of FIG. **1** to form a larger ionic thruster. The arrangement **800** shows the ionic thrusters **102** in a hexagonal (e.g., honeycomb) pattern. In other examples, the arrangement **800** can include a different number of ionic thrusters **102** in a different pattern. The arrangement **800** allows for a larger thrust to be generated using similar ionic thrusters **102**. In this way, the same ionic thrusters **102** and/or thruster components can be used to propel aircraft with larger thrust requirements. Similarly, larger thrust can be developed without requiring the voltage source to provide a different (e.g., larger) voltage to the thruster.

[0031] FIG. **9** is a flowchart describing an example method **900** of generating and controlling thrust for an aircraft using an ionic thruster. In some examples, the method **900** can be performed by a controller of the aircraft (e.g., the controller **110**). The example method **900** begins at block **902**, at which the controller directs a voltage source (e.g., the voltage source **108**) to provide a voltage to electrodes of an ionic thruster (e.g., the pins **210** and the ground electrode **204** of the ionic thruster **102**). The method **900** continues to block **904** where the provided voltage generates an electric field between the electrodes. The electric field strength is based on the magnitude of voltage that is supplied to the electrodes. The method **900** continues to block **906** where the electric field generates a corona discharge (e.g., corona discharge **400**) between the electrodes. Once the electric field strength reaches a threshold level, a fluid (e.g., the fluid **404**, air, etc.) between the electrodes ionizes and a corona discharge forms. The method **900** continues to block **908** where the corona

discharge causes ionic wind (e.g., ionic wind **402**) to generate thrust within the nozzle (e.g., nozzle **200** of ionic thruster **102**). The charged particles of the corona discharge collide with neutral particles of the fluid in the ion thruster to create ionic wind, which results in a generation of thrust. The method **900** continues to block **910**, at which the controller determines whether to change the thrust using voltage. If not, the method **900** moves to block **914**. Otherwise, the method **900** proceeds to block **912** where the voltage provided to the electrodes is changed by the controller. A higher voltage produces a stronger corona discharge (e.g., more charged particles are produced), whereas a lower voltage produces a weaker corona discharge (e.g., fewer charged particles are produced). Therefore, the ionic wind and resultant thrust will change in response to a change in the amount of voltage applied. The method **900** continues to block **914**, at which the controller determines whether to change the thrust using an electromagnet (e.g., electromagnet **500**). If not, the method **900** moves to block **918**. Otherwise, the method **900** proceeds to block **916** where the controller directs the electromagnet to apply a magnetic field to the corona discharge. The magnetic field causes the charged particles of the corona discharge to narrow (e.g., move closer together) and increase speed exiting the ionic thruster. The magnitude of the magnetic pressure varies based on a strength of the magnetic field and, therefore, power supplied to the electromagnet. Thus, thrust can be increased based on the power supplied to the electromagnet. The method **900** continues to block **918**, at which the controller determines whether to continue generating thrust. If so, the method **900** returns to block **908**. Otherwise, the method **900** ends.

[0032] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

[0033] As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more”, and “at least one” are used interchangeably herein.

Furthermore, although individually listed, a plurality of means, elements, or actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0034] As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part

between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.

[0035] As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.

[0036] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

[0037] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly within the context of the discussion (e.g., within a claim) in which the elements might, for example, otherwise share a same name.

[0038] As used herein, “approximately” and “about” modify their subjects/values to recognize the potential presence of variations that occur in real world applications. For example, “approximately” and “about” may modify dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections as will be understood by persons of ordinary skill in the art. For example, “approximately” and “about” may indicate such dimensions may be within a tolerance range of $\pm 10\%$ unless otherwise specified herein.

[0039] From the foregoing, it will be appreciated that example systems, apparatus, articles of manufacture, and methods have been disclosed that generate thrust for aircraft. Further examples and combinations thereof include the following:

[0040] Example 1 includes a thruster for aircraft including a nozzle. The nozzle includes an outlet and an inlet, the inlet to receive fluid and containing an electrode mount. A ground electrode is disposed within the nozzle. Conducting pins are coupled to the electrode mount, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end, the pins spaced apart from the ground electrode.

[0041] Example 2 includes the thruster as recited in example 1, further including a voltage source coupled to the conducting pins and the ground electrode, the voltage source to create an electric field between the pins and the ground electrode.

[0042] Example 3 includes the thruster as recited in example 2, wherein the electric field is to generate a corona discharge.

[0043] Example 4 includes the thruster as recited in example 3, further including an electromagnet, the electromagnet to surround a space between the pins and the ground electrode, the electromagnet to direct the corona discharge towards a central axis of the nozzle, the central axis to extend between the inlet and the outlet.

[0044] Example 5 includes the thruster as recited in example 1, wherein the nozzle is composed of a non-conductive material.

[0045] Example 6 includes the thruster as recited in example 1, wherein the conducting pins are

parallel to a central axis of the nozzle, the central axis to extend between the inlet and the outlet.

[0046] Example 7 includes the thruster as recited in example 1, wherein the electrode mount is composed of a conductive material.

[0047] Example 8 includes the thruster as recited in example 1, further including a dielectric guide having holes therethrough, the holes to surround the pins and allow fluid to flow from the inlet to the outlet.

[0048] Example 9 includes the thruster as recited in example 8, wherein each hole converges towards an end of a respective one of the pins.

[0049] Example 10 includes the thruster as recited in example 1, wherein the ground electrode is a plate having holes to allow fluid to flow between the inlet and the outlet.

[0050] Example 11 includes the thruster as recited in example 1, wherein the electrode mount includes radial supports extending away from a center of the electrode mount to an internal wall of the nozzle.

[0051] Example 12 includes the thruster as recited in example 1, wherein the nozzle converges between the inlet and the outlet.

[0052] Example 13 includes an aircraft including a voltage source and a thruster. The thruster includes a body including an outlet and an inlet, the body to receive fluid and containing an electrode mount, a ground electrode disposed within the body and electrically coupled to the voltage source, and conducting pins coupled to the electrode mount and electrically coupled to the voltage source, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end.

[0053] Example 14 includes the aircraft as recited in example 13, wherein the voltage source is to cause a corona discharge between the conducting pins and the ground electrode.

[0054] Example 15 includes the aircraft as recited in example 14, further including an electromagnet surrounding the respective second ends of the conducting pins and the ground electrode, the electromagnet to narrow the corona discharge between the pins and the ground electrode.

[0055] Example 16 includes the aircraft as recited in example 13, wherein the body is composed of a non-conductive material.

[0056] Example 17 includes the aircraft as recited in example 13, further including a dielectric guide having holes therethrough and surrounding the pins to allow fluid to flow through the dielectric guide.

[0057] Example 18 includes a method for generating thrust on an aircraft. The method includes providing a voltage to a thruster, the thruster including a nozzle including an outlet and an inlet, the inlet to receive air, an electrode mount disposed within the inlet, a ground electrode disposed within the nozzle, and conducting pins coupled to the electrode mount, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end, the pins spaced apart from the ground electrode. The method includes generating a corona discharge with the voltage, the corona discharge to extend between the conducting pins and the ground electrode, and inducing an ionic wind with the corona discharge, the ionic wind to generate the thrust.

[0058] Example 19 includes the method as recited in example 18, wherein a magnitude of the thrust changes in response to a change in the voltage.

[0059] Example 20 includes the method as recited in example 18, the method further including providing a magnetic field around the corona discharge, the magnetic field to be generated by an electromagnet, the magnetic field to affect the corona discharge to increase the thrust.

[0060] The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, apparatus, articles of manufacture, and methods have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary,

this patent covers all systems, apparatus, articles of manufacture, and methods fairly falling within the scope of the claims of this patent.

Claims

- 1.** A thruster for aircraft comprising: a nozzle including an outlet and an inlet, the inlet to receive fluid and containing an electrode mount; a ground electrode disposed within the nozzle; and conducting pins coupled to the electrode mount, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end, the pins spaced apart from the ground electrode.
- 2.** The thruster as recited in claim 1, further including a voltage source coupled to the conducting pins and the ground electrode, the voltage source to create an electric field between the pins and the ground electrode.
- 3.** The thruster as recited in claim 2, wherein the electric field is to generate a corona discharge.
- 4.** The thruster as recited in claim 3, further including an electromagnet, the electromagnet to surround a space between the pins and the ground electrode, the electromagnet to direct the corona discharge towards a central axis of the nozzle, the central axis to extend between the inlet and the outlet.
- 5.** The thruster as recited in claim 1, wherein the nozzle is composed of a non-conductive material.
- 6.** The thruster as recited in claim 1, wherein the conducting pins are parallel to a central axis of the nozzle, the central axis to extend between the inlet and the outlet.
- 7.** The thruster as recited in claim 1, wherein the electrode mount is composed of a conductive material.
- 8.** The thruster as recited in claim 1, further including a dielectric guide having holes therethrough, the holes to surround the pins and allow fluid to flow from the inlet to the outlet.
- 9.** The thruster as recited in claim 8, wherein each hole converges towards an end of a respective one of the pins.
- 10.** The thruster as recited in claim 1, wherein the ground electrode is a plate having holes to allow fluid to flow between the inlet and the outlet.
- 11.** The thruster as recited in claim 1, wherein the electrode mount includes radial supports extending away from a center of the electrode mount to an internal wall of the nozzle.
- 12.** The thruster as recited in claim 1, wherein the nozzle converges between the inlet and the outlet.
- 13.** An aircraft comprising: a voltage source; and a thruster including: a body including an outlet and an inlet, the body to receive fluid and containing an electrode mount; a ground electrode disposed within the body and electrically coupled to the voltage source; and conducting pins coupled to the electrode mount and electrically coupled to the voltage source, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end.
- 14.** The aircraft as recited in claim 13, wherein the voltage source is to cause a corona discharge between the conducting pins and the ground electrode.
- 15.** The aircraft as recited in claim 14, further including an electromagnet surrounding the respective second ends of the conducting pins and the ground electrode, the electromagnet to narrow the corona discharge between the pins and the ground electrode.
- 16.** The aircraft as recited in claim 13, wherein the body is composed of a non-conductive material.
- 17.** The aircraft as recited in claim 13, further including a dielectric guide having holes therethrough and surrounding the pins to allow fluid to flow through the dielectric guide.
- 18.** A method for generating thrust on an aircraft, the method comprising: providing a voltage to a thruster, the thruster including: a nozzle including an outlet and an inlet, the inlet to receive air; an electrode mount disposed within the inlet; a ground electrode disposed within the nozzle; and

conducting pins coupled to the electrode mount, each of the pins having a first end coupled to the electrode mount and a second end positioned closer to the ground electrode than the first end, the pins spaced apart from the ground electrode; generating a corona discharge with the voltage, the corona discharge to extend between the conducting pins and the ground electrode; and inducing an ionic wind with the corona discharge, the ionic wind to generate the thrust.

19. The method as recited in claim 18, wherein a magnitude of the thrust changes in response to a change in the voltage.

20. The method as recited in claim 18, the method further including: providing a magnetic field around the corona discharge, the magnetic field to be generated by an electromagnet, the magnetic field to affect the corona discharge to increase the thrust.
